Module 1, Part 2: Predicate Logic and Quantifiers

Discrete Mathematics (Course Code: 3341)

1 The Necessity of Predicate Logic

While propositional logic deals with fixed statements that are either true or false, it is limited in its ability to express general mathematical properties that apply to groups of objects. For instance, it cannot easily represent a statement like "All prime numbers are odd except 2". **Predicate logic** extends propositional logic by introducing variables and predicates, allowing us to describe properties shared by many objects.

2 Fundamental Definitions

- **Predicate:** A statement of the form $P(x)$, where x is the **subject variable** and P is the **property** assigned to that variable.
- **Propositional Function:** The expression $P(x)$ is not a proposition on its own because it lacks a truth value until a specific value is assigned to x or it is bound by a quantifier.
- **Universe of Discourse (UD) / Domain:** The set of all possible values that the variable x can take.
- **Free vs. Bound Variables:** A variable is **free** if it can take any value from the UD without restriction. It becomes **bound** once it is constrained by a quantifier like $\forall$ or $\exists$.
- **Well-Formed Formula (WFF):** A syntactically correct logical expression constructed using predicates, variables, connectives, and quantifiers.

3 Quantifiers

Quantifiers are symbols used to indicate the extent to which a predicate property applies over a specific domain.

3.1 The Universal Quantifier ($\forall$)

- **Symbol and Meaning:** Denoted by $\forall$ (an inverted A), it stands for "**for all**," "**for every**," or "**for each**".
- **Truth Conditions:**
  - **TRUE:** If $P(x)$ is true for **EVERY** x in the UD.
  - **FALSE:** If there is **AT LEAST ONE** x where $P(x)$ is false. This specific x is known as a **counterexample**.

3.2 The Existential Quantifier ($\exists$)

- **Symbol and Meaning:** Denoted by $\exists$ (a backward E), it stands for "**there exists**," "**for some**," or "**for at least one**".
- **Truth Conditions:**
  - **TRUE:** If $P(x)$ is true for **AT LEAST ONE** x in the UD.
  - **FALSE:** If $P(x)$ is false for **EVERY SINGLE** x in the UD.

3.3 The Uniqueness Quantifier ($\exists!$)

This "third useful quantifier" indicates that there exists **exactly one** (unique) element in the UD such that $P(x)$ is true.

4 Negating Quantified Statements (De Morgan's Laws)

To negate a statement with a quantifier, swap the quantifier type and negate the inner predicate property.

1. **Negating Universal:** $\neg(\forall x P(x)) \equiv \exists x \neg P(x)$
   Logic: "Not all x satisfy P" is equivalent to "There exists at least one x that does not satisfy P".
2. **Negating Existential:** $\neg(\exists x P(x)) \equiv \forall x \neg P(x)$
   Logic: "There is no x satisfying P" is equivalent to "Every x fails to satisfy P".

5 Workout Problems and Examples

Problem 1: Symbolic Translation
Translate the following into symbolic form:

- **"All apples are green."**
  Solution: Let $P(x)$ be "x is green." If the UD is the set of apples: $\forall x P(x)$.
- **"Some chalkboards are black."**
  Solution: Let $b(x)$ be "x is black." If the UD is chalkboards: $\exists x b(x)$.

Problem 2: Truth Value Determination
Let the UD be all real numbers. Determine the truth value of:

1. $\forall x(x^2 \geq 0)$: **True**. The square of every real number is non-negative.
2. $\forall x(|x| > 0)$: **False**. The **counterexample** is $x = 0$, because $|0| = 0$, which is not greater than 0.

Problem 3: Complex Negation
Negate the statement: **"For every real number x, if $x > 2$, then $x^2 > 4$."**

1. **Identify:** Let $P(x)$ be "$x > 2$" and $Q(x)$ be "$x^2 > 4$."
2. **Symbolic Form:** $\forall x(P(x) \to Q(x))$.
3. **Negate:** $\neg[\forall x(P(x) \to Q(x))] \equiv \exists x \neg(P(x) \to Q(x))$.
4. **Apply Implication Law:** Since $\neg(P \to Q) \equiv P \wedge \neg Q$, the result is $\exists x(P(x) \wedge \neg Q(x))$.
5. **Translation:** "There exists a real number x such that $x > 2$ and $x^2 \leq 4$."

6 Rules of Inference for Quantified Statements

These rules are used to add or remove quantifiers during formal proofs:

1. **Universal Instantiation (UI):** If $\forall x P(x)$ is true, then $P(c)$ is true for **any** element c in the domain.
2. **Universal Generalization (UG):** If $P(c)$ is true for an **arbitrary** element c, then $\forall x P(x)$ is true.
3. **Existential Instantiation (EI):** If $\exists x P(x)$ is true, then $P(c)$ is true for **some specific** (new) element c.
4. **Existential Generalization (EG):** If $P(c)$ is true for a **known** element c, then $\exists x P(x)$ is true.